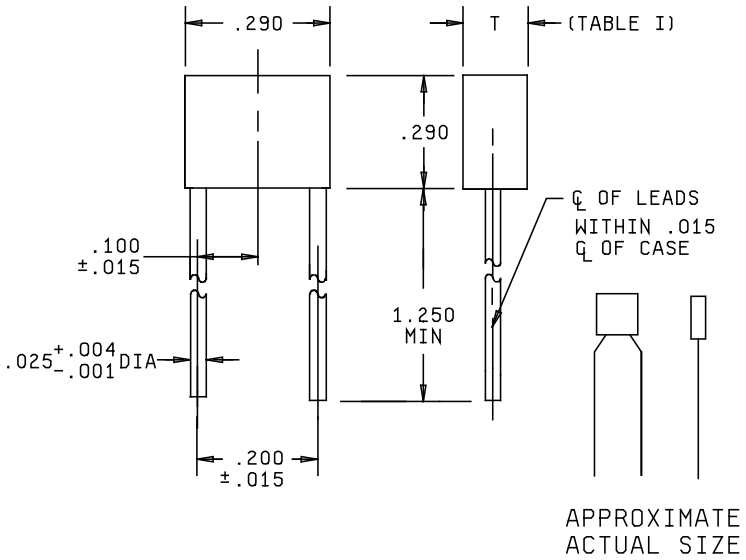
INCH-POUND
MIL-PRF-39014/20F
8 June 2021
SUPERSEDING
MIL-PRF-39014/20E
3 November 2005

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLE CKR08

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39014.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .004 | 0.10 |
| .015 | 0.38 |
| .025 | 0.64 |
| .100 | 2.54 |
| .200 | 5.08 |
| .290 | 7.37 |
| 1.250 | 31.75 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Unless otherwise specified, tolerance is ± .010 inch (0.25 mm).
4. Insulating coating shall not extend more than .010 inch (0.25 mm) along lead wires.

FIGURE 1. Capacitor style CKR08.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Molded or preformed.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage:

Body insulation: Shall be able to withstand 1,300 volts between leads and case.

Insulation resistance:

At +25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.

At +125°C: Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-PRF-39014.

| Capacitance change with reference to +25°C | |
|---|---|
| Steps A through D of MIL-PRF-39014 table VI Bias = 0 volts | Steps E through G of MIL-PRF-39014 table VI Bias = rated voltage |
| ±15 percent | +15, -25 percent |

Immersion:

IR: Not less than 50 percent of initial +25°C requirements.

Salt atmosphere (corrosion): In accordance with MIL-PRF-39014.

Resistance to soldering heat:

ΔC: +15, -5 percent of the initial +25°C measurement.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

Life:

Rated conditions, 100 percent of dc rated voltage at 125°C.

0 through 3,000 hours:

IR: At +25°C, not less than 50 percent of initial +25°C requirement.

At +125°C, not less than 50 percent of initial +125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

4,000 through 32,000 hours:

IR: At +25°C, not less than 15 percent of initial +25°C requirement.

At +125°C, not less than 15 percent of initial +125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

Accelerated conditions, 200 percent of dc rated voltage at 125°C.

0 and 250 hours:

IR: At +25°C, not less than 50 percent of initial +25°C requirement.

At +125°C, not less than 50 percent of initial +125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

1,000, 2,000 and 4,000 hours:

IR: At +25°C, not less than 15 percent of initial +25°C requirement.

At +125°C, not less than 15 percent of initial +125°C requirement.

ΔC: ±20 percent from initial measured value.

Dissipation factor: Not greater than 2.5 percent.

Part or identifying number (PIN): M39014/20- (dash number from table I).



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics and dash number.

| Dash number 1/ | | | | Capacitance, pF | Capacitance tolerance, ± percent | DC rated voltage, volts | Dimension, inches 2/ | |
|---|---|---|---|---|---|---|---|---|
| Failure rate level (%/1,000 hours) | | | | | | | | |
| M (1.0) | P (0.1) | R (0.01) | S (0.001) | | | | | |
| 0004 | 0007 | 0010 | 0013 | 1,200,000 | 10 | 50 | .130 (3.30) | ±.010 (.25) |
| 0005 | 0008 | 0011 | 0014 | 1,500,000 | 10 | 50 | .130 (3.30) | ±.010 (.25) |
| 0006 | 0009 | 0012 | 0015 | 2,000,000 | 10 | 50 | .130 (3.30) | ±.010 (.25) |
| 0104 | 0107 | 0110 | 0113 | 1,200,000 | 10 | 50 | .145 (3.68) | ±.015 (.38) |
| 0105 | 0108 | 0111 | 0114 | 1,500,000 | 10 | 50 | .145 (3.68) | ±.015 (.38) |
| 0106 | 0109 | 0112 | 0115 | 2,000,000 | 10 | 50 | .145 (3.68) | ±.015 (.38) |

1/ Complete PIN is in the format M39014/20-NNNN, where NNNN is the 4 digit dash number from the table.
2/ Metric equivalents are given in parentheses.

TABLE II. Substitutability data.

| Dash numbers in MIL-PRF-39014/20F 1/ | | | | Substitutable for dash numbers in: MIL-C-39014/20A(EC) | | | | |
|---|---|---|---|---|---|---|---|---|
| M | P | R | S | L | M | P | R | S |
| 0004 | 0007 | 0010 | 0013 | 0001 | 0004 | 0007 | 0010 | 0013 |
| 0005 | 0008 | 0011 | 0014 | 0002 | 0005 | 0008 | 0011 | 0014 |
| 0006 | 0009 | 0012 | 0015 | 0003 | 0006 | 0009 | 0012 | 0015 |
| 0104 | 0107 | 0110 | 0113 | 0101 | 0104 | 0107 | 0110 | 0113 |
| 0105 | 0108 | 0111 | 0114 | 0102 | 0105 | 0108 | 0111 | 0114 |
| 0106 | 0109 | 0112 | 0115 | 0103 | 0106 | 0109 | 0112 | 0115 |

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all higher FR level columns.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2021-018)

Review activities:
Navy - AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.